MILITARY SPECIFICATION

INTEGRATED ELECTRONIC CENTRAL

AN/ASQ-88(*)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. The Electronic Central covered by this specification shall provide communication, navigation and identification facilities required in naval aircraft to carry out military missions.

1.2 Classification. This central shall consist of the items listed in 6.9.

1.3 Design Data. No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (See paragraph 6.7).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in 6.11, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein: (see 6.5(4)).

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Parts and Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL–E–5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL–E–5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4).

(4) When previously produced models of this central did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 *Nonstandard Parts and Materials Approval.* Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL–E–5400.

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3.2.2 *Electron Devices.* Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL–E–4682.

3.2.3 *Microelectronic Modular Assemblies.* When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency. Microelectronic Modular Assemblies (Microelectronic Circuits) (MICS) shall be contained within Maintenance Modules (for example, on printed circuit cards). The choice of Microelectronic Modular Assemblies shall be based on at least two sources of supply, in accordance with WR–98.

3.2.4 *Modular Construction.* The use of modular space assignment and construction of easily disconnected plug-in subunits or component assemblies shall be in accordance with Specification MIL–E–19600 where possible except that for the KY–532A/ASQ and RT–736/ASQ–88 ease of maintenance shall have primary consideration. The following considerations shall apply where practicable.

(1) Modules shall be of standardized form and dimensions to facilitate use in other equipment of different over-all configuration and shall be designed for maximum practical effectiveness of cooling air supplied by or to the equipment.

(2) Each module shall contain the circuitry of one or more functions of the central to facilitate testing and repair. The central shall be divided into as many modular units as is electrically and physically practical in keeping with efficient space utilization, equipment performance and reliability.

(3) It shall be possible to operate the central with one or more modules removed from the chassis but connected to it electrically by means of one or more extension cable assemblies and by use of mechanical extenders when required.

(4) It shall be possible to replace any module with other modules of the same type and obtain specified equipment performance with only minor readjustments.

3.3 Design and Construction. The central shall conform with all the applicable requirements of Specification MIL–E–5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 *Total Weight.* The total weight of the central excluding cables shall be a minimum consistent with good design and shall not exceed 227.5 pounds.

3.3.2 *Reliability.*

3.3.2.1 *Operational Stability.* The central shall operate with satisfactory performance, continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of controls which are inaccessible to the operator during normal use.

3.3.2.2 *Operating Life.* The central shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 *Time Totalizing Meter.* A time totalizing meter conforming, as applicable, with Specification MIL–M–7793 and MS17322 shall be included in the following:

RT–546/ASQ–19
RT–736/ASQ–88
KY–532A/ASQ
KY–531/ASQ–88
AM–2349/ASQ–19

3.3.2.4 *Reliability in Mean Time Between Failures (MTBF).* The central shall have 50 hours of mean (operating) time between

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failures when tested and accepted as outlined under the requirements of 4.4.3. For individual items when purchased separately the requirements shall be as shown in 6.9.

3.3.3 *Cabling and Connections.*

3.3.3.1 *Cables and Connectors.* The central shall provide for the use of cables and connectors in accordance with Specification MIL–E–5400.

3.3.3.2 *Interconnection Cabling.* The central shall be capable of satisfactory operating using external wiring in accordance with the applicable requirements of Specification MIL–W–5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the central.

3.3.4 *Control Panels.* All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL–C–6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 *Interchangeability of Reordered Equipment.* For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 *Interference Control.* The generation of radio interference by the central and the vulnerability of the central to radio interference shall be controlled within the limits of Specification MIL–I–6181 except as specified below and in the individual equipment specification.

3.3.6.1 *Short Duration.* Short Duration interference shall not exceed the limits of Figures 3 through 5.

3.3.7 *Maintenance Provisions and Field Testing.* Provisions for maintenance shall be as specified in Specification MIL–E–5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 *Nomenclature and Nameplates.* Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL–N–18307.

3.3.9 *Standard Conditions.* The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 volts, 380–420 cps, 3ϕ
	26 ± 1.0 volts, 380–420 cps, 1ϕ
	27.5 ± 0.5 volts dc
System Cooling Air	157.5 lb per hour

3.3.10 *Service Conditions.* The central shall operate satisfactorily under any of the environmental conditions or reasonable combination of these conditions as specified in Specification MIL–E–5400 for the Classes of Equipment specified below unless otherwise specified herein or in the individual equipmnt specifications:

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Receiver-Transmitter, Radar	RT–736/ASQ–88 Class 2
Decoder, Pulse	KY–531/ASQ–88 Class 2
Coder-Receiver-Transmitter	KY–532A/ASQ Class 2
Amplifier-Power Supply-Receiver	AM–2349/ASQ–19 Class 2
Antenna	AS–909/ARA–48 Class 2
Control, Central	C–6248/ASQ–88 Class 1
Indicator, Frequency Channel	ID–808/ASQ Class 1
Receiver-Transmitter, Radio	RT–546A/ASQ–19 See 3.3.10.3
Intercommunication Station	LS–460/AIC Class 1

3.3.10.1 *System Cooling Air Requirements.* Cooling air requirements for the central, when the equipments are mounted on the central plenum shelf, are given in Figure 2 of this specification for high line voltages.

3.3.10.2 *Vibration.* The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL–E–5400.

3.3.10.3 *Pressure, Temperature and Altitude.* The RT–546A shall operate satisfactorily when subjected to the environmental temperatures of Figure 1 and at a maximum pressure altitude of 25,000 feet. The RT–546A shall also be capable of operating satisfactorily for 10 minutes at a pressure altitude of 40,000 feet and at temperatures of —10° to 130°F under emergency conditions.

3.3.11 *Warm-Up Time.* The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.3.12 *Primary Input Power Requirements.* The central shall meet all applicable requirements of MIL–STD–704 unless otherwise specified herein and shall give specified performance from the following power sources with characteristics as defined in MIL–STD–704 having limits as specified therein unless otherwise specified below. The power required shall not exceed the specified amounts.

(1) A–C Power (Three-Phase) 115/200 V, Category "B", 685 VA

(2) A–C Power (Single-Phase) 26 V, Within Limits Proportional to Category "B", 135 VA

(3) D–C Power 28–V, Category "B", 4 amps, see 6.10

(Additional current requirements not to exceed 8 amperes per equipment while channeling.)

3.3.13 *Associated Equipment.* This equipment shall operate with the associated equipment listed in 6.9.1.

3.4 Detail Requirements.

3.4.1 *Communication Function.* The communication function shall be performed by Receiver-Transmitter, Radio RT–546A/ASQ–19 in accordance with Specification MIL–R–22762.

3.4.2 *TACAN Function.* The navigation function shall be performed by the Receiver-Transmitter, Radar RT–736/ASQ–88 and the Decoder, Pulse KY–531/ASQ–88 in accordance with Specification MIL–R–81105.

3.4.3 *Auxiliary Receiver, Power Supply and ADF Function.* The auxiliary receiver, power supply and automatic direction finder function shall be performed by Amplifier-Power Supply-Receiver AM–2349/ASQ–19 and Antenna AS–909/ARA–48 in accordance with Specification MIL–A–22765.

3.4.4 *IFF/Coder Function.* The IFF/Coder function shall be performed by Coder-Receiver-Transmitter KY 532A/ASQ in accord-

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ance with Specification MIL–C–22764.

3.4.5 *Central Control.* The remote control functions required for the operation of the communication receiver, auxiliary receiver. and UHF navigation function shall be performed by Control, Central C–6248/ASQ–88 in accordance with Specification MIL–C–22766.

3.4.6 *Channel Indicator.* The channel indicator function for the remote indication of the communication receiver shall be performed by Indicator, Frequency Channel ID–808/ASQ Specification in accordance with MIL–C–7953.

3.4.7 *Intercommunication Stations.* The forward and aft intercommunication station functions shall be performed by Intercommunication Station LS–460/AIC in accordance with Specification MIL–I–22767.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Classification of Tests.* Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests. Preproduction tests shall be made on one or more individual equipments and centrals representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 *Preproduction Test Data.* The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 *Scope of Tests.* Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL–T–5422 and interference tests in accordance with Specification MIL–I– 6181. Cooling air shall be supplied to the central and 3ϕ 115VAC, 400 cps power shall be applied to equipment internal blower at all times during steps 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, and 14 of the temperature altitude test as outlined in MIL–T–5422. Cooling air may be applied at least 30 minutes prior to applying any other power to equipments when application is required during or following the humidity test specified in MIL–T–5422.

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4.2.3 *Preproduction Approval.* Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 *Production Equipments.* Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests. One of the first 10 production centrals shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 *Scope of Tests.* This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 *Accessory Material.* In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 *Initial Production Sample Approval.* Approval of the Initial Production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 *Reconditioning of Initial Production Test Sample.* On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests

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(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 *Individual Tests.* Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run In Test

4.4.1.1 *Examination of Product.* Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 *Operational Test.* Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 *Manufacturing Run In Test.* Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes or rotary for 10 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 *Sampling Tests.* Centrals selected for sampling tests shall first have passed the individual tests. Centrals shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of centrals offered for acceptance	Quantity of centrals selected for sampling tests
First 10	1
Next 50	1
	One for each additional 100 or fraction thereof

4.4.2.1 *Scope of Tests.* As a minimum, each central selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all

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necessary measurements to assure that all applicable specification requirements have been met.

(2) Operation test at certain environmental conditions. The conditions may vary for each central tested and should be based on results of the preproduction and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with not restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 *Reliability Assurance Tests.* Unless otherwise stipulated in the contract, equipments which have successfully passed the individual tests shall be tested for reliability assurance in accordance with MIL–STD–781.

4.4.3.1 *Test Level.* The test Level shall be Level G of MIL–STD–781.

4.4.3.2 *Accept-Reject Criteria.* Figure III of MIL–STD–781 shall be used to determine the accept-reject criteria.

4.4.3.3 *Length of Heat portion of Cycle.* After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

4.4.3.4 *Performance Characteristics to be Measured.*

(1) *UHF Communications — RT–546A*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Receiver Sensitivity	Standard 5 uv/10 db Service 20 uv/10 db	20 uv/10 db
Audio Capability	Standard not less than 250 mw Service not less than 62.5 mw	Not less than 62.5 mw
RF Power Output	Standard not less than 16 w Service not less than 9 w	Not less than 9 w
Modulation Capability (input to clipping)	Not less than 70%	Not less than 70%
Tone Modulation Capability	NLT 70%	NLT 70%
Side Tone Level	Standard not less than 250 mw Service not less than 75 mw	There shall be side tone output

(2) *Receiver/ADF/Power Supply — AM–2349 and AS–909*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Receiver Sensitivity	Standard 6 uv/10 db Service 18 uv/10 db	18 uv/10 db
Audio Capability	Standard not less than 250 mw Service not less than 62.5 mw	Not less than 62.5 mw
Power Supply Voltages		
430 VDC	370 to 505	360 to 515
275 VDC	241 to 302	230 to 310
250 VDC	218 to 278	210 to 290
130 VDC	114 to 142	110 to 145
140 VDC	137 to 143	135 to 145
–180 VDC	–148 to –207	–140 to –215

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ADF RF Operational Test	Not more than 6 sec. for antenna to assume new heading 180° removed from previous heading	Not more than 12 sec.

(3) *Intercommunication Sets — LS-460*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Power Output	Standard not less than 275 mw	Not less than 137 mw

(4) *TACAN — KY-531 and RT-736*

Characteristic	*Specification Requirement*	*Acceptance Requirement (if lower)*
Distance Accuracy and Sensitivity	Standard ± (0.2 mi + 0.1%) for sig levels from –117 to –30 dbw Service ± (0.3 mi + .1%) at –114 dbw to –30 dbw	± (0.3 mi + 0.1%) for sig levels from –114 to –30 dbw with 40% modulation (20% 15 cycles, 20% 135 cycles)
Bearing Accuracy and Sensitivity	Standard ± 2° with sig levels from –113 to –107 dbw and –35 to –30 dbw ± 1° with signal level from –107 to –35 dbw Service ±2.5° with sig levels from –110 to –104 dbw and –35 to –30 dbw ± 1.5° with signal levels from –104 to –35 dbw	± 2.5° with sig levels from –110 to –30 dbw with 40% modulation (20% 15 cycles, 20% 135 cycles)
Interrogation Coding T/R	12 ± 0.5 usec	12 ± 0.5 usec
Identification	Standard not less than 50 mw audio for sig levels from –117 to –30 dbw; 30db S/N ratio from –114 to –30 dbw. Service not less than 50 mw audio for signal levels from –114 to –30 dbw; 30 db S/N ratio from –111 to –30 dbw.	Not less than 50 mw audio with 30 db S/N ratio for sig level from –111 to –30 dbw
		Not less than 27 dbw
Power Output	T/R Standard not less than 30 dbw Service not less than 27 dbw	Not less than 24 dbw
	A/A Standard not less than 27 dbw Service not less than 24 dbw	Not less than 24 dbw

(5) *IFF/Coder — KY-532A/ASQ*

IFF/Coder performance characteristics to be measured on the KY-532A shall be in accordance with specification MIL-C-22764.

4.4.3.5 *Failure Criteria.* Performance requirements below those specified herein (4.4.3.4) shall constitute one failure unless otherwise specified in the approved detailed test procedures (see 4.6). If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused other parts to fail.

4.4.3.6 *Determination of Compliance.* Compliance shall be determined in accordance with Figure 13 of Specification MIL-R-23094 (see paragraph 3.19 therein).

4.4.4 *Special Tests.* Special tests shall be conducted on a quantity of equipments for purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipments selected for special tests may be selected from equipments previously subjected to the sampling test.

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4.4.4.1 *Special Tests Schedule.* Selection of individual equipments and centrals for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.4.2 *Scope of Tests.* Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 *Equipment Failure.* Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 500-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 500 hours each of test time.

Quantity of equipments offered for acceptance	Quantity to be selected for life test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 *Test Conditions.* The central life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0–5000 ft.)
Humidity	Room ambient
A–C Voltage	115 ± 5 volts (at lowest applicable frequency)
D–C Voltage	27.0 ± 2.0 volts
Cooling Air	157.5lb per hour

4.5.2 *Test Periods.* The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the central shall be turned on and off several times and put through its various phases of operation.

4.5.3 *Performance Check.* At approximately 8-hour intervals during the test, a limited

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performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 *Test Data.* The contractor shall keep a daily record of the performance of the central, making particular note of any deficiencies of failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part Number

(2) The circuit reference symbol-number

(3) The part function

(4) Name of manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 *Failure Report.* In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 *Reconditioning of Life Test Samples.* A central which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the central shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the central for acceptance.

4.6 **Test Procedures.** The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL–T–18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, such procedures may be used when approved by the procuring activity. However, the right is reserved to require modification of such procedures, including additional tests, when deemed necessary.

4.7 **Reconditioning of Tested Equipment.** Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 **Presubmission Testing.** No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 **Rejection and Retest.** Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 **General.** All major units and parts of

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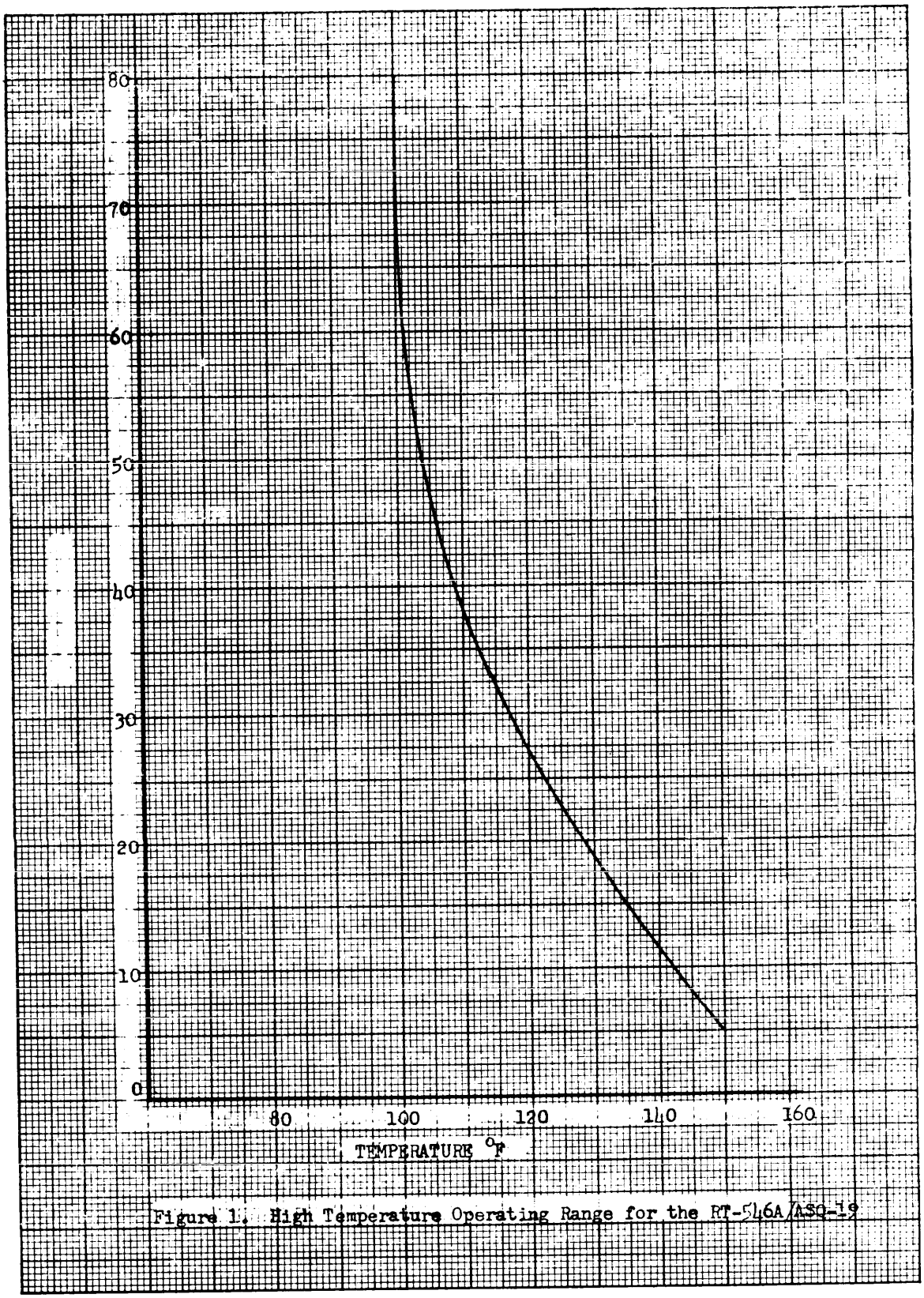


Figure 1. High Temperature Operating Range for the RT-546A/ASQ-19

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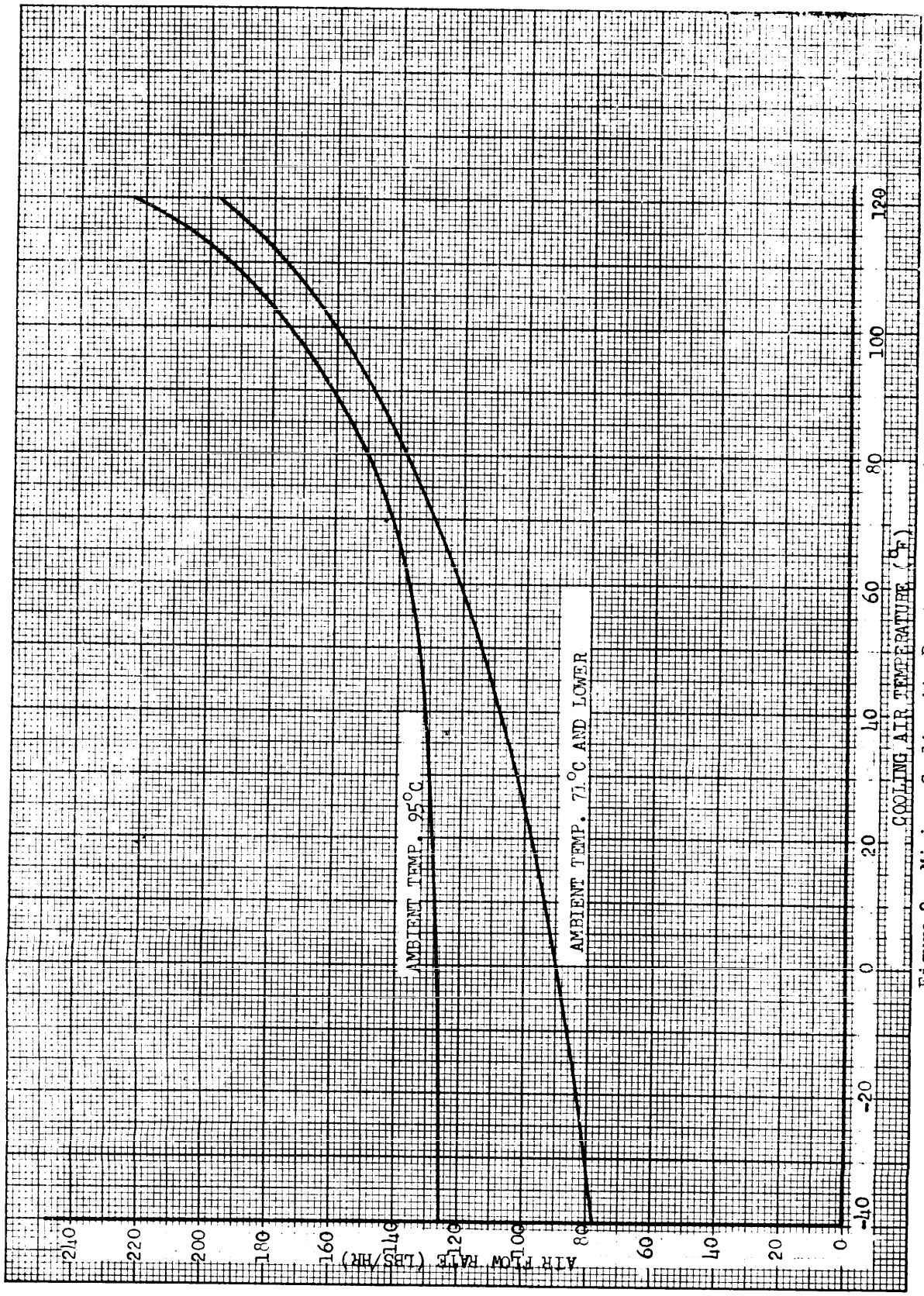


Figure 2. Minimum Cooling Air Requirements - AN/ASQ-88

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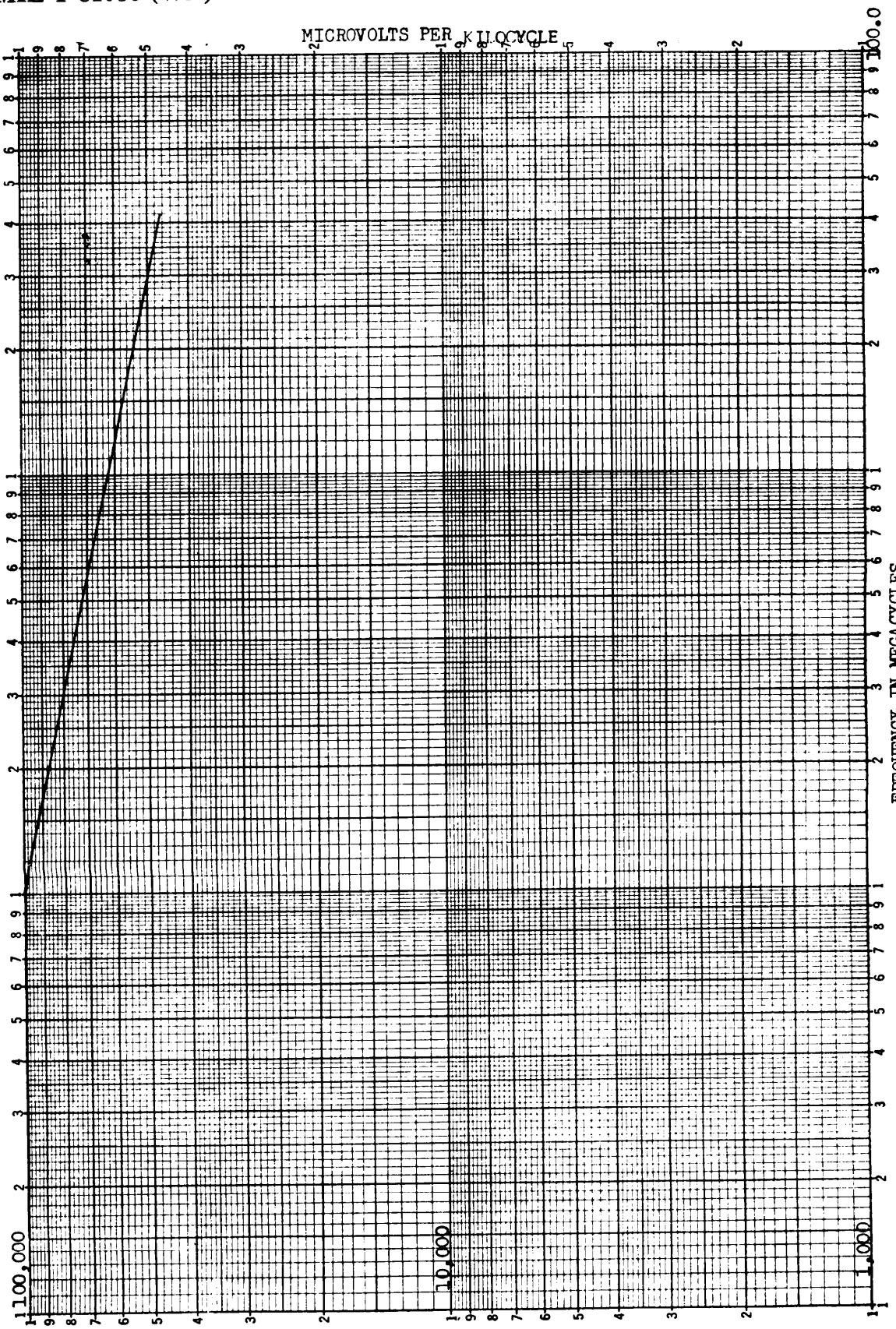


Figure 3. Conducted Interference Limits - Broadband

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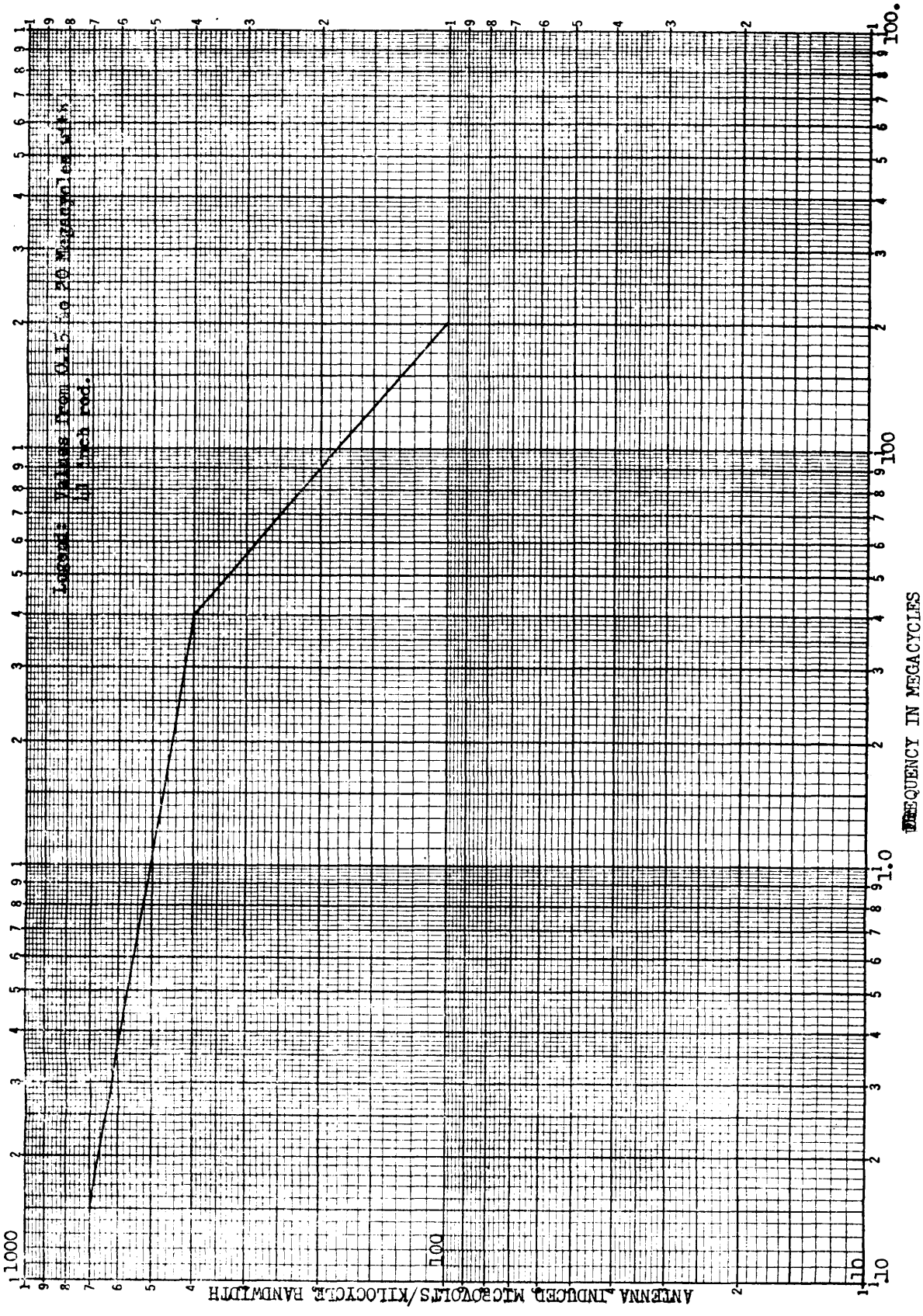


Figure 4. Standard Limits Broadband Radiated Interference

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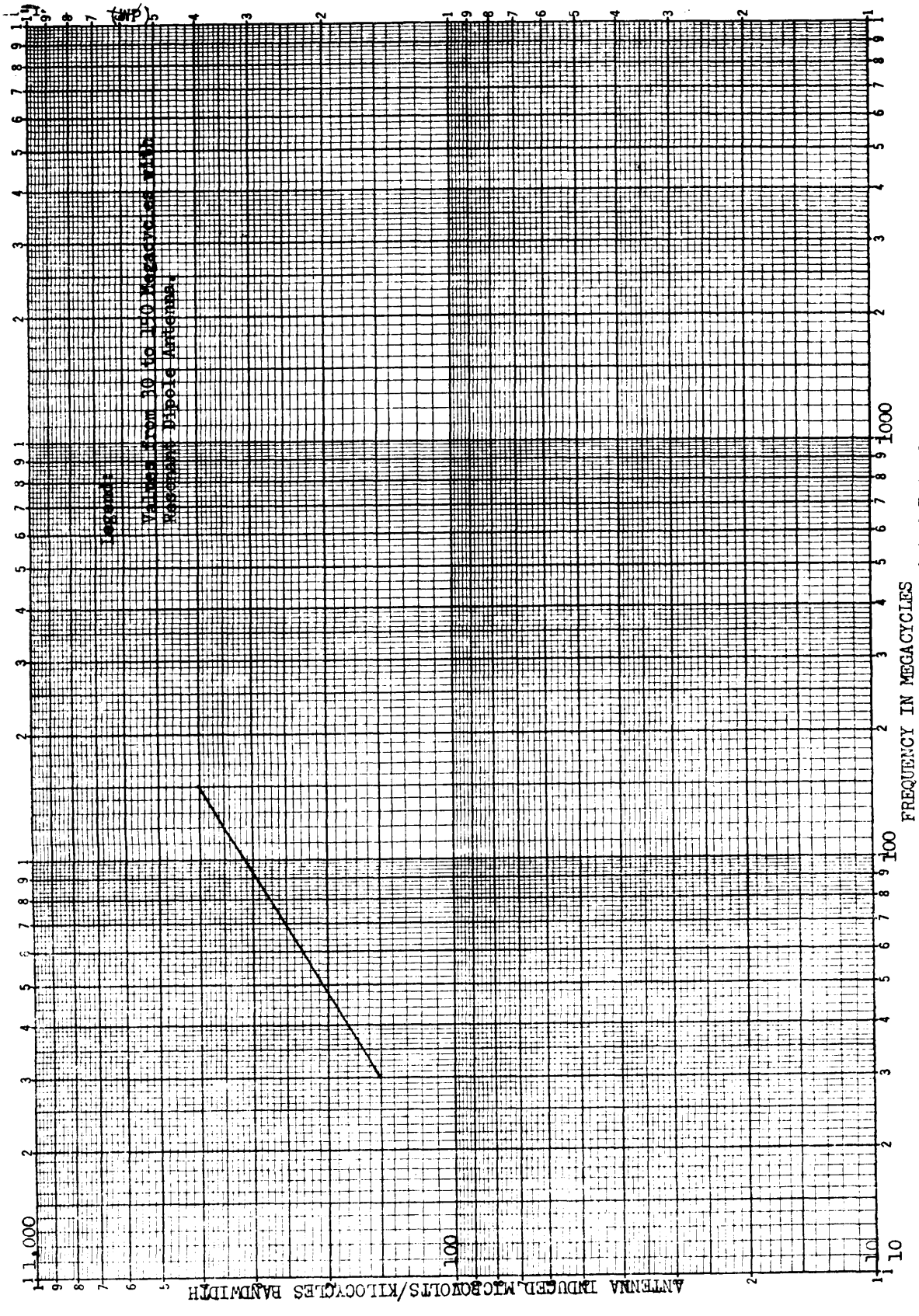


Figure 5. Standard Limits Broadband Radiated Interference

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the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL–E–17555.

6. NOTES

6.1 Intended Use. The electronic central covered by this specification shall provide communication, navigation, and identification required in naval aircraft to carry out military missions.

6.2 Test Values. Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 v dc, 26 ± 1.0 v ac and 115 ± 1.0 v ac as applicable. These data are to be used in testing the equipments and at installation points for compliance with minimum acceptable standards of performance.

6.3 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL–E–5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) *Contract.* The contract shall have precedence over any specification.

(2) *This Specification.* This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) *Referenced Specifications.* Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(4) *Reordered Equipment.* The following specifications shall take precedence over those concerning performance requirements listed or referenced in paragraph 2.1 where equipment is recorded.

MIL–I–6181B—shall be the governing specification with the following exception:

1. For the tests required under paragraph 3.1.1 (MIL–I–6181B), Figures 3–5 (MIL–I–81036) shall be used in lieu of Figures 1, 16, and 17. (MIL–I–6181B) for equipment other than the KY–532A/ASQ. Performance of the other equipment in the Integrated Elec-

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tronic Central shall not degrade the performance of the KY-532A/ASQ which shall comply with Military Specification MIL-C-22764.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 1.3).

6.8 Asterisk. In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.9 Items of Equipment. The central covered by this specification shall consist of the following items:

Item	Appl. para.	BTBF	Type designation	MIL-R-23094 accept/ reject figure
Receiver-Transmitter, Radio	3.4.1	170	RT-546A/A5Q-19	13
Receiver-Transmitter, Radar	3.4.2	200	RT-736(*)/ASQ-88	13
Decoder, Pulse			KY-531(*)/ASQ-88	13
Coder-Receiver-Transmitter	3.4.4	325	KY-532A/ASQ	14
Amplifier-Power-Supply-Receiver	3.4.3	400	AM-2349(*)/ASQ-19	14
Antenna	3.4.3	2500	AS-909(*)/ARA-48	16
Control, Central (2)	3.4.5	900	C-6248(*)/ASQ	15
Intercommunication Station (2)	3.4.7	4000	LS-460(*)/AIC	16
Indicator, Frequency Channel (2)	3.4.6	2700	ID-808(*)/ASQ	16

6.9.1 *Associated Equipment.* The equipment shall operate with the following associated equipment:

Item	Type designation
Indicator, Bearing-Distance-Heading	ID-663/U
Control, Transponder Set	C-6280(P)/APX

6.10 Power Requirements. The D-C Power Requirements shall be as specified in 3.3.12 and MIL-STD-704 except that the lower voltage limits under emergency conditions shall be 21 volts in lieu of 17 volts.

6.11 Applicable Documents.

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SPECIFICATIONS

MILITARY

MIL-E-4682 — Electron Tubes and Transistors, Choice and Application of

MIL-W-5088 — Wiring; Aircraft, Installation of

MIL-E-5400 — Electronic Equipment, Aircraft, General Specification for

MIL-T-5422 — Testing, Environmental Aircraft Electronic Equipment

MIL-I-6181 — Interference Control Requirements, Aircraft Equipment

MIL-C-6781 — Control Panel: Aircraft Equipment, Rack or Console Mounted

MIL-M-7793 — Meter, Time Totalizing

MIL-E-17555 — Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-T-18303 — Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

MIL-N-18307 — Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-E-19600 — Electronic Modules, General Specification for

MIL-I-22075 — Indicator, Bearing-Distance - Heading ID-663/U

MIL-R-81105 — Receiver - Transmitter, Radar RT-736/ASQ-88 and Decoder, Pulse KY-531/ASQ-88

MIL-C-22764 — Coder - Receiver - Transmitter KY-532/ASQ

MIL-A-22765 — Amplifier-Power Supply-Receiver AM-2349/ASQ-19 and Antenna AS-909/ARA-48

MIL-I-22767 — Intercommunication Station LS-460/AIC and Intercommunication Station LS-459/AIC

MIL-I-22768 — Indicator, Frequency Channel ID-808/ASQ

MIL-C-38257 — Control, Transponder Set. C-6280(P)/APX

MIL-R-22762 — Receiver - Transmitter, Radio RT-546A/ASQ-19

MIL-C-22766 — Control, Center C-6248/ASQ-88

STANDARDS

MILITARY

MIL-STD-704 — Electric Power, Aircraft Characteristics and Utilization of

MS17322 — Meter, Time Totalizing, 115 Volt, 400-cycle, Rev. B

MIL-STD-781 — Reliability Tests, Exponential Distribution

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BUREAU OF NAVAL WEAPONS

WR–98 — Microelectronic Devices used in Avionic Equipment, procedures for

6.11.1 *Availability of Documents.*

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.12 Criteria for Multiple Source. The criteria of this paragraph shall be used for determining the extent to which the requirement for multiple independent source availability of microcircuits is satisfied. For interchangeability at the MIC level, multiple source availability exists if the devices from the different sources may be used interchangeably, device by device, in the final equipment without any need for mechanical or electrical changes. For interchangeability at the card level, multiple source availability exists if a card composed of MICS from one source is mechanically and electrically interchangeable with another card composed of MICS from another source. Numerous aspects need be considered to arrive at the independence of sources; these involve production independence, corporate (or firm) and financial independence, and geographical separation. The production independence of one source or another is determined by the extent to which the one source is independent of the other for manufacture; and by the lack of common original suppliers of raw or processed materials. Design, process information, technology, and diffusion masks may be derived from a common source. However, in such a case, each device source shall have a demonstrated capability or originating these items. The corporate (or firm) and financial independence of one source of another is determined by the absence of common financial interests. Geographical separation is simply the distance between the locations where principal manufacture takes place by two sources.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81036(WP) Integrated Electronic Central AN/ASQ-88(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

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Washington, D. C. 20360

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